

Analog VLSI (EEE/INSTR F313): last-minute formulas

One pack per topic: the idea in plain words, the formulas to memorise (how to say them and how to check them), and every question type with the paper and question where it was asked.

1. MOSFET bias: the DC recipe
2. Small signal, gain and the impedance rules
3. Differential pair
4. Five-transistor OTA and its bandwidth
5. Performance parameters: gain error, settling, slewing
6. One-stage op amps: telescopic, buffer window
7. Design procedure and scaling
8. Folded cascode
9. Two-stage op amp
10. Gain boosting
11. Common-mode feedback
12. Slew rate and input range
13. Stability: phase margin and peaking
14. Frequency compensation: dominant pole and Miller

LAST-MINUTE PACK

MOSFET bias: the DC recipe

The idea in plain words

Every question starts with DC. **Assume saturation**, use the square law to link I_D and V_{ov} , **walk the node voltages** from a rail (node = supply – drops), then **check the fence**. Going the other way (design), pick V_{ov} , get W/L from the square law. PMOS: same equations with magnitudes.

Memorise these

Formula	Say it as	Check it with
$V_{ov} = V_{GS} - V_{th}$	overdrive is how far the gate is above threshold	typically 0.1–0.5 V
$I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} V_{ov}^2$	current grows with the square of the overdrive	double $V_{ov} \rightarrow 4\times$ current
$\frac{W}{L} = \frac{2I_D}{\mu C_{ox} V_{ov}^2}$	square law backwards	answers 5–2000 are normal
$V_D \geq V_G - V_{th}$ (NMOS), $V_D \leq V_G + V_{th} $ (PMOS)	the fence: drain no lower (NMOS) / no higher (PMOS) than one V_{th} past the gate	every swing and CM limit is one of these
$I_D = \mu C_{ox} \frac{W}{L} \left[V_{ov} V_{DS} - \frac{V_{DS}^2}{2} \right]$	triode: below the fence	deep triode $\approx$ resistor $1/(\mu C_{ox}(W/L)V_{ov})$

Memory line: “Assume, solve, walk, check.”

Every question type, and how to solve it

If the question...	Do this	Where it was asked
Find ID, VD for a given bias	$V_{ov} = V_{GS} - V_{th} \rightarrow I_D$ by square law $\rightarrow V_D = V_{DD} - I_D \cdot R_D \rightarrow$ check the fence.	Worked Example 1 (analyse a common-source stage) Worked Example 3 (a PMOS with magnitudes)
Size a device for a current / voltage	Pick or read $V_{ov} \rightarrow W/L = 2I_D/(\mu C_{ox} \cdot V_{ov}^2) \rightarrow \text{gate} = V_{th} + V_{ov}$.	Worked Example 2 (design a biased NMOS) Tutorial 1 Q1 (design a pair with a mirror tail)

Small signal, gain and the impedance rules

The idea in plain words

Gain is always $A_v = -G_m R_{out}$. G_m is the input device’s g_m (divided down by any source degeneration). R_{out} is everything at the output node in parallel, found with the **three impedance rules**: into a gate ∞ , into a drain r_O (multiplied by $g_m \cdot r_O$ for each device stacked under it), into a source $1/g_m$. Cascodes multiply; diodes are $1/g_m$; the smallest resistance in parallel wins.

Memorise these

Formula	Say it as	Check it with
$g_m = \frac{2I_D}{V_{ov}} = \sqrt{2\mu C_{ox} \frac{W}{L} I_D}$	g_m three ways	mS for 0.1–1 mA
$r_O = \frac{1}{\lambda I_D}$	output resistance is 1 over lambda I	tens–hundreds of kΩ
$A_v = -G_m R_{out}$	gain is G_m times R_{out} , sign by inspection	CS inverts; follower and CG don’t
$R_{up/down} = g_m r_O \cdot r_{O,below}$	up multiplies, down divides — by $g_m \cdot r_O$	cascode $\approx g_m \cdot r_O^2$ (MΩ)
$R_{into\ source} \approx 1/g_m, \quad R_{diode} = 1/g_m$	looking into a source or a diode: $1/g_m$	kΩ
$ A_v = \frac{R_{output}}{R_{source\ path}}$ (transistor = $1/g_m$)	ratio rule for degeneration	$g_m \cdot R_D / (1 + g_m \cdot R_S)$

Memory line: “Gate ∞ , drain r_O , source $1/g_m$. Up multiplies, down divides.”

Every question type, and how to solve it

If the question...	Do this	Where it was asked
Gain of any one-device stage	STEP B roles → STEP C each load’s resistance → STEP D $A_v = -G_m \cdot R_{out}$.	Worked Example 1 (analyse a common-source stage)

If the question...	Do this	Where it was asked
Cascode / telescopic output resistance	$R_{down} = g_m \cdot r_O \cdot r_O$ (NMOS stack), $R_{up} = g_m \cdot r_O \cdot r_O$ (PMOS stack); $R_{out} = R_{up} \parallel R_{down}$. A simple load throws the cascode away (load trap).	Lab 3 (CS versus cascode gain, bandwidth and GBW) Lab 5 (simple versus low-compliance cascode mirror)

Differential pair

The idea in plain words

Split inputs into **CM** (average) and **DM** (difference). DM: the tail node is an AC ground, use the **half circuit** — $A_d = g_m(R_D \parallel r_O)$. CM: the tail splits into **2RSS** per half — $A_{CM} = -R_D / (1/g_m + 2R_{SS})$. The input CM range is two fences: the tail plus VGS1 at the bottom, M1 against its drain at the top.

Memorise these

Formula	Say it as	Check it with
$A_d = g_m(R_D \parallel r_O)$	half circuit = one CS stage	current-source load: $g_m \cdot r_O/2$
$A_{CM} = -\frac{R_D}{1/g_m + 2R_{SS}}$	CM half circuit has twice the tail	$\ll 1$
$V_{in,CM} : [V_{ISS} + V_{GS1}, V_{D1} + V_{th}]$	CM range: tail + VGS up to drain + Vth	floor < ceiling
$\Delta V_{in,full} = \sqrt{2} V_{ov}$	the pair fully steers at $\sqrt{2} \cdot V_{ov}$	used in slewing

Memory line: “CM sees 2RSS, DM sees a ground.”

Every question type, and how to solve it

If the question...	Do this	Where it was asked
Design a pair: RD, W/L, R, CM range	ID = ISS/2 → RD from the drop → W/L by square law → CM floor/ceiling fences.	Tutorial 1 Q1 (design a pair with a mirror tail)
Diode or current-source loads: Ad	Half circuit; diode load = $1/g_{m3}$ (gain = $g_{m1}/g_{m3} = \sqrt{(\mu_n(W/L)1/\mu_p(W/L)3)}$); current source = r_O .	Tutorial 1 Q2 (pair with diode-connected PMOS loads) Tutorial 1 Q3 (pair with PMOS current-source loads)
Resistive tail: VCM, ACM, CM rise to triode	VCM = VGS + ISS·RSS; Ad = $g_m \cdot R_D$; ACM = $-R_D / (1/g_m + 2R_{SS})$; $\Delta V_{CM} = (V_D - V_{CM} + V_{th}) / (1 - A_{CM})$.	Tutorial 1 Q4 (a single-supply pair with an RSS tail)

Five-transistor OTA and its bandwidth

The idea in plain words

The mirror adds the two half-currents, so $G_m = g_{m1}$ and $R_{out} = r_{O2} \parallel r_{O4}$. Four exam numbers come out of one picture: the gain, the input CM range (tail + VGS1 up to VDD – |VGS3| + Vth), the output swing (one overdrive at each rail plus the tail), and one pole at the output. In unity feedback the output resistance falls to $1/g_{m2}$ and the pole jumps to g_{m2}/C_L (= GBW).

Memorise these

Formula	Say it as	Check it with
$A_v = g_{m1}(r_{O2} \parallel r_{O4})$	gm times rO over two	30–200
$V_{in,CM,min} = V_{ov5} + V_{GS1}$, $V_{in,CM,max} = V_{DD} - V_{GS3} + V_{thn}$	floor: tail + VGS; ceiling: the diode costs a full VGS3	given CM limits → solve backwards for sizes
$\text{swing} = (V_{DD} - V_{ov4}) - (V_{ov5} + V_{ov2})$	output rails minus overdrives	$\approx V_{DD} - 0.5 \text{ V}$
$f_{-3dB} = \frac{1}{2\pi(r_{O2} \parallel r_{O4})C_L}$, $GBW = \frac{g_{m1}}{2\pi C_L}$	one pole at the output; GBW = gm/CL	GBW = $A_v \times f_{-3dB}$
$f_{buffer} = \frac{g_{m2}}{2\pi C_L}$, $SR = \frac{I_{SS}}{C_L}$	buffer bandwidth = GBW; slew = tail current over CL	MHz; V/μs
$P = V_{DD}(I_{ref} + I_{SS})$	power counts every branch	μW–mW

Memory line: “The mirror recovers the lost half.”

Every question type, and how to solve it

If the question...	Do this	Where it was asked
Given sizes → gain, swing, CM range, power	Mirror ratio → ISS → ID → gm, rO ($\lambda \approx 1/L!$) → the four formulas.	Quiz 1 2024-25 Q1 (five-transistor OTA from its sizes) Problem Set 1 P1 (five-transistor OTA, full analysis)

If the question...	Do this	Where it was asked
Given CM limits → sizes (mid-sem style)	Tail Vov5 from its size → floor gives VGS1 → (W/L)1; ceiling gives VGS3 → (W/L)3; then gain, swing, f−3dB, buffer f.	Mid-sem exam Q1 (five-transistor OTA (a–e)) Quiz 1 Part A (five-transistor OTA) Quiz 1 Part B (five-transistor OTA)
Given SR and CL → current, BW, GBW	ISS = SR·CL → mirror for Iref → Rout → f−3dB; GBW = gm/(2πCL).	Quiz 1 2024-25 Q2 (the same OTA sized for 10 V/μs into 2 pF)

Performance parameters: gain error, settling, slewing

The idea in plain words

Feedback trades a huge, sloppy open-loop gain A for an accurate $1/\beta$. The **error** is $\varepsilon = 1/(1 + \beta A)$. A one-pole op amp in feedback settles like an RC with $\tau = 1/(\beta\omega_u)$; to reach error ε takes $\ln(1/\varepsilon)$ time constants. A big step first **slews** at $SR = I/C_L$, then settles linearly.

Memorise these

Formula	Say it as	Check it with
$A_{closed} = \frac{A}{1 + \beta A} \approx \frac{1}{\beta}, \beta = \frac{R_2}{R_1 + R_2}$	closed-loop gain ≈ 1 over beta	gain 10 $\rightarrow \beta = 0.1$
$\varepsilon = \frac{1}{1 + \beta A} \approx \frac{1}{\beta A}, A_{min} = \frac{A_{closed}}{\varepsilon}$	error is 1 over loop gain	1% at gain 10 $\rightarrow A \geq 1000$
$\omega_u = A_0\omega_0, \tau = \frac{1}{\beta\omega_u}$	GBW fixed; $\tau = 1/(\beta\omega_u)$	ns for GHz
$t_s = \tau \ln \frac{1}{\varepsilon}$	1% = 4.6 τ , 0.1% = 6.9 τ	$\ln 100, \ln 1000$
$\tau_{cl} = \frac{R_{out}C_L}{1 + \beta A}, SR = \frac{I}{C_L}$	feedback speeds the output node up by $(1 + \beta A)$	initial slope = V/τ

Memory line: “Error is one over the loop gain; settling is $\ln(1/\varepsilon)$ time constants.”

Every question type, and how to solve it

If the question...	Do this	Where it was asked
Gain-error spec $\rightarrow$ minimum A	$\beta = 1/A_{closed} \rightarrow A \geq A_{closed}/\varepsilon.$	Razavi Example 9.1 (how much open-loop gain for 1% gain error?) Problem Set 1 P2 (gain error and settling of the P1 op amp)
Settling-time spec $\rightarrow$ minimum ω_u / f_u	$t = \ln(1/\varepsilon)/(\beta\omega_u) \rightarrow \text{solve } \omega_u \rightarrow f_u = \omega_u/2\pi.$	Razavi Example 9.2 (how fast must the op amp be to settle in 5 ns?)

If the question...	Do this	Where it was asked
Closed loop with Rout and CL: τ, $V_{out}(t)$, slewing	$ACL = A/(1 + \beta A)$; $\tau = R_{outCL}/(1 + \beta A)$; $V_{out} = V_0 \cdot ACL(1 - e^{(-t/\tau)})$; slews if $V_0 \cdot ACL/\tau > SR$.	Tutorial 6 Q1 (linear settling versus slewing)

One-stage op amps: telescopic, buffer window

The idea in plain words

Telescopic cascode: gain $g_{m1}(R_{up} \parallel R_{down}) \approx (g_m r_O)^2/2$, swing = VDD minus the whole stack of overdrives (twice, differentially). A cascode-mirror load costs one extra $|V_{thp}|$. Used as a **buffer** (output tied to an input gate) two fences fight and leave a window only $V_{th} - V_{ov4}$ wide.

Memorise these

Formula	Say it as	Check it with
$A = g_{m1}(g_{m3}r_{O3}r_{O1} \parallel g_{m5}r_{O5}r_{O7})$	gm times cascode cascode	1000s
$\text{swing}_{diff} = 2[V_{DD} - \sum V_{ov} - V_{ISS}]$	twice what the stack leaves	each device costs its $ V_{ov} $
$V_{b1} - V_{th4} \leq V_{out} \leq V_{b1} - V_{GS4} + V_{th2}$	buffer window, width $V_{th} - V_{ov4}$	$\approx 0.5\text{ V}$
$V_{out,max} = V_b + V_{thp} $ (PMOS cascode), $V_{out,min} = 2V_{ov} + V_{thn}$ (NMOS cascode mirror)	cascode fences	mirror stack costs a V_{th}

Memory line: “Cascode buys gain with swing; a buffer gets only $V_{th} - V_{ov}$.”

Every question type, and how to solve it

If the question...	Do this	Where it was asked
Telescopic: gain, swing, bias voltages	Currents $\rightarrow$ gm, rO each side $\rightarrow$ R_{up} , R_{down} $\rightarrow$ gain; swing from the stack; V_b 's put devices at their edges.	Problem Set 1 P3 (telescopic gain, swing and bias) Mid-sem 2024-25 Q3 (PMOS-input telescopic with a cascode mirror, and as a buffer) Tutorial 2 Q2 (telescopic with a diode-connected cascode mirror)

If the question...	Do this	Where it was asked
Output tied to an input (buffer): Vout range	Write M4's fence (floor) and M2's fence with its gate = Vout (ceiling or floor).	Tutorial 3 Q1 (telescopic with a low-voltage cascode mirror) Mid-sem 2024-25 Q3 (PMOS-input telescopic with a cascode mirror, and as a buffer) Buffer window lecture (the telescopic buffer window) Problem Set 1 P5 (mirror-loaded telescopic and its buffer window)
Fully differential pair with current-source loads	$A_d = g_{m1}(r_{O1} \parallel r_{O3})$; each output from $V_{in,CM} - V_{th}$ up to $V_{DD} - V_{ov3} $.	Tutorial 2 Q1 (fully differential pair with PMOS current-source loads)

Design procedure and scaling

The idea in plain words

One recipe for every design: **power** → **currents**, **swing** → **overdrives**, **square law** → **W/L**, **check the gain**, and if it is short, lengthen the weak side ($\times 2$ W and L keeps V_{ov} , halves λ). Then set each bias voltage so its device sits at its edge. Linear scaling ($W, I \times \alpha$) keeps every voltage and the gain, multiplies g_m and power by α .

Memorise these

Formula	Say it as	Check it with
$I = P/V_{DD}$	power budget	mA
$\sum V_{ov} = V_{DD} - \text{swing per side}$	swing budget	tail gets the most
$g_m r_O \propto \sqrt{WL/I_D}$, $\lambda \propto 1/L$	longer devices, more intrinsic gain	$L \times 2 \rightarrow r_O \times 2$
$I_{SS} = SR \cdot C_L$	slew spec sets the tail	μA
$W, I \times \alpha$: $g_m \times \alpha$, $r_O \div \alpha$, V_{ov}, A_v same	linear scaling	$\alpha = CL_{\text{new}}/CL_{\text{old}}$

Memory line: “Power, swing, V_{ov} , W/L, gain, lengthen.”

Every question type, and how to solve it

If the question...	Do this	Where it was asked
Design a telescopic for power, swing, gain	I from power; split overdrives; W/L each; $R_{up} \parallel R_{down}$; lengthen PMOS if short; $V_{in,CM}$, V_{b1} , V_{b2} .	Razavi Example 9.7 (design a telescopic op amp from specs) Tutorial 2 Q3 (design a folded cascode for 2.4 V swing and 6 mW) Problem Set 1 P6 (design a folded cascode (2.0 V swing, 4.5 mW))
High-swing telescopic from SR and V_{out} limits	$I_{SS} = SR \cdot C_L$; floor = 2 NMOS V_{ov} ; ceiling = $V_{DD} - (\sqrt{2} + 2) V_{ov} $ (tail carries 2ID); W/L; V_b 's; gain.	Mid-sem 2023-24 Q3 (design a high-swing PMOS-input telescopic)

If the question...	Do this	Where it was asked
Scale for a bigger load	$\alpha = \text{CL ratio; power} \times \alpha; \text{widths} \times \alpha.$	Razavi Example 9.8 / Problem Set 1 P9 (linear scaling to drive a bigger load)

Folded cascode

The idea in plain words

Fold the input pair sideways into the cascodes: the bottom sources carry $I_{SS}/2 + I$. $R_{out} = R_{up} \parallel R_{down}$ with the input device's r_O in parallel at the fold node; $G_m \approx g_{m1}$ (current divider). The big win is the **input CM range**: a PMOS input can go below ground ($V_{ov} - |V_{thp}|$), an NMOS input above VDD.

Memorise these

Formula	Say it as	Check it with
$I_{D,bottom} = \frac{I_{SS}}{2} + I$	the fold node carries both	currents add
$R_{up} = g_m r_O r_O, R_{down} = g_m r_O (r_{O1} \parallel r_{O,src})$	the fold node has r_{O1} in parallel	$R_{down} < R_{up}$ usually
$G_m \approx g_{m1}$	almost all of M1's current goes up the cascode	fraction $\approx 0.9-1$
$V_{in,CM}^{PMOS} : [V_{ov} - V_{thp} , V_{DD} - V_{ov,tail} - V_{GS1}]$	PMOS input: CM below ground	negative floor
$V_{in,CM}^{NMOS} : [V_{ov,tail} + V_{GS1}, V_Y + V_{th1}]$	NMOS input: CM above VDD possible	cap at VDD

Memory line: "Folding flips the CM inequality."

Every question type, and how to solve it

If the question...	Do this	Where it was asked
Folded cascode: CM range, bias limits, swing, gain	Fold node $Y = V_{b2} + V_{GS3} $; V_b limits put sources at their edges; output from $V_{b1} - V_{th}$ to $V_{b2} + V_{th} $; gain $g_m(R_{up} \parallel R_{down})$.	Quiz 1 2023-24 Q2 (NMOS-input folded cascode: CM range, bias limits, swing, gain) Problem Set 1 P8 (folded cascode CM ranges (input = output CM is possible))
How much of M1's current reaches the output	Divider: $(r_{O1} \parallel r_{O5}) / [(1/g_{m3} \parallel r_{O3}) + (r_{O1} \parallel r_{O5})]$.	Problem Set 1 P7 (how much of M1's current reaches the output?)

If the question...	Do this	Where it was asked
Design a folded cascode	Same design recipe; four overdrives share the room; PMOS input → CM can be 0 V.	Tutorial 2 Q3 (design a folded cascode for 2.4 V swing and 6 mW) Problem Set 1 P6 (design a folded cascode (2.0 V swing, 4.5 mW))

Two-stage op amp

The idea in plain words

One stage for gain, one for swing: $A = A_1 A_2$. The first-stage output CM is not free — it must sit where the second-stage device carries its current ($X = V_{DD} - |V_{GS}|$), and that caps the input CM. Two high-impedance nodes mean two poles, hence compensation (L13–L14).

Memorise these

Formula	Say it as	Check it with
$A = A_1 A_2, A_2 = g_{m,2nd}(r_O \parallel r_O)$	gains multiply	1000s–10000s
$V_X = V_{DD} - V_{GS,2nd} $	X is pinned by the second stage	input CM $\leq X + V_{th}$
$V_{pp,diff} = 2(V_{DD} - V_{ov} - V_{ov})$	CS output: one overdrive at each rail	nearly rail to rail

Memory line: “High gain, then high swing.”

Every question type, and how to solve it

If the question...	Do this	Where it was asked
Two-stage: CM at X, Y; gain; swing	X from the second stage’s VGS; $A_1 \times A_2$; swing from the CS stage.	Tutorial 3 Q2 (two-stage op amp, CM level at X and Y) Tutorial 3 Q3 (two-stage with a telescopic first stage)

Gain boosting

The idea in plain words

Gm is hard to raise; **Rout is not**. An auxiliary amplifier A1 holds the cascode's source still, so it fights back $(1 + A_1)$ times harder: $R_{out} \approx (1 + A_1)g_{m2}r_{O2}r_{O1}$, gain $\approx (g_m r_O)^3$. Three ways to build A1 (Lec 8): a CS stage (costs headroom: $V_{out,min} = V_{GS3} + V_{ov2}$), a PMOS CS (needs $V_{GS2} \leq |V_{th3}|$), a folded cascode (gain $g_m(R_{up,aux} \parallel R_{down,aux})$). The load must be boosted too, or its rO decides the gain.

Memorise these

Formula	Say it as	Check it with
$R_{out} = R_S + r_O + (1 + A_1)g_m R_S r_O$	test-source result (Lec 7)	$A_1 = 0 \rightarrow$ plain degeneration
$R_{out} \approx (1 + A_1)g_{m2}r_{O2}r_{O1}$, $G_m \approx g_{m1}$	boosted cascode	GΩ–TΩ possible
$A_1 = g_{m3}r_{O3}$ (CS), $A_1 = g_m(R_{up,aux} \parallel R_{down,aux})$ (folded)	the auxiliary's own Gm·Rout	10s–1000s
$V_X = V_{GS3}$, $V_{G2} = V_X + V_{GS2}$, $V_{out,min} = V_{GS3} + V_{ov2}$	CS auxiliary bias and its headroom cost	walk up from ground
$A_v = -g_{m1}(R_{up} \parallel R_{down})$	with a real PMOS load, Rup = its rO wins	load trap

Memory line: “Hold the source still and the cascode fights $(1 + A_1)$ times harder.”

Every question type, and how to solve it

If the question...	Do this	Where it was asked
CS-auxiliary boosted cascode: bias, swing, gain	$V_X = V_{GS3}$ at I1; $V_P = V_X + V_{GS2}$ at I2; swing $[V_{GS3} + V_{ov2}, V_{DD} - V_{ov,I2}]$; $A_1 = g_{m3}(r_{O3} \parallel r_{O,I1})$; gain $= -g_{m1}(r_{O,I2} \parallel A_1 g_{m2} r_{O2} r_{O1})$.	Mid-sem 2025-26 Q1 (gain-boosted cascode with a CS auxiliary) Tutorial 4 Q1 (regulated cascode with an NMOS auxiliary)

If the question...	Do this	Where it was asked
Folded-cascode auxiliary: Aaux, Rout, Vb limits	$A_{aux} = g_{m,in}[g_{m,rO} \parallel g_{m,rO}(r_O \parallel r_O)] \rightarrow R_{out} = A_{aux} \cdot g_{m2} r_{O2} r_{O1}$; $V_{b2,min}$ puts the sink at its edge.	Mid-sem 2024-25 Q4 (R_{out} of a gain-boosted cascode with a folded auxiliary) Tutorial 4 Q3 (gain boosting with a folded-cascode auxiliary)
PMOS auxiliary: region checks	PMOS fence on the auxiliary: its drain (M_2 's gate) $\leq V_P + V_{thp} \rightarrow V_{GS2} \leq V_{th3} $.	Tutorial 4 Q2 (gain boosting with a PMOS auxiliary)

Common-mode feedback

The idea in plain words

A fully differential output sits between two current sources, so its CM **floats**. CMFB **senses** (resistors, followers or triode devices), **compares** with VREF and **corrects** a current source. Triode sensing turns $V_{out1} + V_{out2}$ into a resistance; the tail current through it fixes V_P , which pins the output CM.

Memorise these

Formula	Say it as	Check it with
$R_{tot} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{out1} + V_{out2} - 2V_{th})}$	two triode devices = one resistor	full triode: $ID = \frac{1}{2}\mu C_{ox}(W/L)[2V_{ov}V_{DS} - V_{DS}^2]$
$V_{out1} + V_{out2} = \frac{2I_D}{\mu_n C_{ox} (W/L) V_P} + 2V_{th}$	the CM equation	$V_P = V_{b1} - V_{GS}$
$A_d = g_m(r_O \parallel r_o \parallel R)$	sensing resistors load the gain	$R \gg r_O$ barely matters
$A_{CM} \approx \frac{r_{O3}}{2r_{O5}}, \quad A_{CM,fb} = \frac{A_{CM}}{1 + T}$	CM gain without and with the loop	CMRR = A_d/ACM
$A_{CM,req} = \frac{2(0.01)V_{O,CM}}{V_{in,CM,max} - V_{in,CM,min}}$	course rule for “±1%”	$V_{in,CM,max} = V_{O,CM} + V_{th}$
$V_{O,CM,opt} = \frac{1}{2} [V_{out,min} + V_{out,max}]$	optimum CM = middle of the swing	VREF = that

Memory line: “Sense, compare, correct.”

Every question type, and how to solve it

If the question...	Do this	Where it was asked
Size triode sensing devices for an output CM	$V_{GS} = V_{out,CM}$, $V_{DS} = V_P$; full triode $I_D = \frac{1}{2}\mu C_{ox}(W/L)[2(V_{GS} - V_{th})V_P - V_P^2]$ per device.	Mid-sem 2025-26 Q4 (size the triode CMFB devices (= Tutorial 5 Q1)) Tutorial 5 Q1 (size the triode CMFB devices) Quiz 2 Part A (triode-sensing CMFB on a telescopic)
Resistive sensing: A_d, ACM, $V_{O,CM}$, CMRR, loop gain	$A_d = g_m(r_O \parallel r_O \parallel R)$; $ACM = r_{O3}/(1/g_m + 2r_{O5})$; $V_{O,CM}$ = middle of the range; ACM_{req} by the $\pm 1\%$ rule; $T = \text{sense} \rightarrow \text{error amp} \rightarrow \text{tail}$; $ACM/(1 + T)$.	Mid-sem 2024-25 Q1 (CMFB with $R = 10\text{ M}\Omega$ sensing (= Tutorial 5 Q3)) Tutorial 5 Q3 (CM gain and CMRR with and without CMFB) Quiz 2 2024-25 Q2 (resistive-sensing CMFB: V_{REF} , $V_{in,CM}$ and the CM gain for $\pm 1\%$)
Which pair for the error amp; CMFB loop gain	Match the error amp's input CM to the sensed level; loop gain = $A_{EA} \cdot g_m \cdot (R_{up} \parallel R_{down})$.	Tutorial 5 Q2 (which pair for the CMFB amplifier, and the loop gain)
Replica CMFB	$M_{14} = M_{11}$, $(W/L)_{15} = (W/L)_{12} + (W/L)_{13}$; conductances match.	Problem Set 2 P5 (the replica CMFB of Lec 12)

Slew rate and input range

The idea in plain words

A big step steers the whole tail to one side: the output ramps at **SR = I/CL** (5-T OTA: ISS/CL; fully differential telescopic: ISS/2CL per output, ISS/CL differentially; folded cascode: limited by IP). Rail-to-rail input uses an NMOS and a PMOS pair in parallel.

Memorise these

Formula	Say it as	Check it with
$SR = \frac{I_{SS}}{C_L}$	slew = current over capacitance	V/μs
$\frac{dV_{out1}}{dt} = \frac{I_{SS}}{2C_L}, \frac{dV_{out,d}}{dt} = \frac{I_{SS}}{C_L}$	fully differential telescopic	each side half
$\Delta V_{in} > \sqrt{2}V_{ov} \Rightarrow$ slewing	full steering	

Memory line: “A fixed tap fills the bucket.”

Every question type, and how to solve it

If the question...	Do this	Where it was asked
5-T OTA: slew, then settle	SR = ISS/CL; slews until the input difference < $\sqrt{2}V_{ov}$; then $\tau = R_{out}CL/(1 + \beta A_0)$.	Tutorial 6 Q2 (a 5-T OTA slews, then settles)
Folded-cascode slew rate	Output = (IP – ID2) – mirrored (IP – ID1); a branch can't go negative; need IP ≥ ISS.	Tutorial 6 Q3 (folded-cascode slew rate)
Telescopic slew	Each output ISS/(2CL); difference ISS/CL.	Problem Set 2 P1 (slew rate of a fully differential telescopic op amp)

Stability: phase margin and peaking

The idea in plain words

The loop gain $\beta A(j\omega)$ must fall below 1 before its phase reaches -180° (Barkhausen). $PM = 180^\circ + \angle \beta A(\omega_{GX})$. Each pole costs up to 90° (45° at the pole). One pole: always stable (PM = 90°). Two close poles with a big gain: tiny PM. Small PM shows up as **peaking**: at ω_{GX} the closed loop is $\frac{1/\beta}{2 \sin(PM/2)}$.

Memorise these

Formula	Say it as	Check it with
$PM = 180^\circ - \sum \tan^{-1} \frac{\omega_{GX}}{\omega_{pi}}$	add the arctangents at crossover	degree mode!
$ \beta A(\omega_{GX}) = 1 : \omega_{GX} \approx \sqrt{\beta A_0 \omega_{p1} \omega_{p2}}$ (above both poles)	crossover	$\beta = 1$ is the worst case
$K = \frac{1}{2 \sin(PM/2)}, PM = 2 \sin^{-1} \frac{1}{2K}$	peaking $\leftrightarrow$ PM	$45^\circ \rightarrow 1.3, 60^\circ \rightarrow 1.0$
$f'_p = f_p(1 + \beta A_0)$	feedback moves one pole up	gain $\times$ BW constant
$Q = \frac{\sqrt{(1 + \beta A_0) \omega_{p1} \omega_{p2}}}{\omega_{p1} + \omega_{p2}}$	two-pole closed loop	coincide Q = 0.5, flat Q = 0.707

Memory line: “Below 1 before -180° .”

Every question type, and how to solve it

If the question...	Do this	Where it was asked
Two poles + gain → PM	Solve $ \beta A = 1$ (Solver or the quadratic in f^2) → $PM = 180^\circ - \sum \text{atan}$.	Mid-sem 2025-26 Q5 (phase margin of two close poles, and PM from 50% peaking) Razavi Problem 10.3 ($A_0 = 1000$ with two close poles) Razavi Problem 10.1 (the largest A_0 for 60° phase margin) Razavi Problem 10.2 (two equal poles, 60° margin, gain 1 and gain 4) Problem Set 2 P2 (phase margin at gain 1 and at gain 5)
Peaking ↔ PM	$K = 1/(2 \sin(PM/2))$ either way.	Mid-sem 2025-26 Q5 (phase margin of two close poles, and PM from 50% peaking) Mid-sem 2023-24 Q5 (peaking factor K at PM 50°; C_c vs C_L for 45°) Razavi Problem 10.4 (50% peaking → what phase margin?) Lecture notes Lec 16 (how much the closed loop peaks at PM = 5°, 45°, 60°) Past tutorial 2024-25 T2 Ex 4 (closed-loop gain at ω_1 for PM 30°, 60°, 90°)

If the question...	Do this	Where it was asked
Feedback moves poles; coincident / maximally flat	One pole: $\times (1 + \beta A_0)$. Two poles: Q formula; $Q = 0.5$ or $0.707 \rightarrow \beta$.	Past tutorial 2024-25 T2 Ex 1 (how far feedback moves a single pole) Past tutorial 2024-25 T2 Ex 2 (two poles closing in — coincident poles and maximally flat) Past tutorial 2024-25 T2 Ex 3 (crossover and PM of a one-pole op amp at gain 100)

Frequency compensation: dominant pole and Miller

The idea in plain words

Make the loop gain cross 0 dB before the second pole: move the first pole down (**dominant pole**) or, in a two-stage op amp, put **Cc across the second stage** — the Miller effect multiplies it by $(1 + A_2)$, splitting the poles. Results: $GBW = g_{m1}/2\pi C_c$, $f_{p2} = g_{m2}/2\pi C_L$, RHP zero $g_{m2}/2\pi C_c$, $SR = I_5/C_c$. $C_c \geq 0.22C_L$ gives 60° when the zero is at $10 \cdot GBW$.

Memorise these

Formula	Say it as	Check it with
$GBW = \frac{g_{m1}}{2\pi C_c}, f_{p2} = \frac{g_{m2}}{2\pi C_L}, f_z = \frac{g_{m2}}{2\pi C_c}$	the three Miller numbers	$f_z > f_{p2}$ when $C_c < C_L$
$SR = \frac{I_5}{C_c}$	slew uses Cc	
$P'_1 \approx \frac{1}{R_1 A_2 C_c}, C_{in} = C_c(1 + A_2)$	Miller pushes the first pole down	
$C_c \geq 0.22C_L$ (60°), $g_{m2} \geq 10g_{m1}$ ($z = 10 GB$)	design rules	45° with z at $10GB \rightarrow 0.122C_L$
$f_D = \frac{f_{gx}}{\beta A_0}$	dominant pole from the crossover you want	-20 dB/dec

Memory line: “Split the poles, mind the zero.”

Every question type, and how to solve it

If the question...	Do this	Where it was asked
Analyse a Miller two-stage: SR, GBW, A0, BW, p2, z, PM	Mirror ratios $\rightarrow$ currents $\rightarrow g_{m1}, g_{m7}, r_{O's} \rightarrow$ the formulas; PM = $90^\circ - \text{atan}(GBW/f_{p2}) - \text{atan}(GBW/f_z)$.	Mid-sem 2024-25 Q2 (analyse a Miller two-stage op amp) Quiz 2 2024-25 Q1 (GBW, PM and Cc from a Miller op amp’s Bode data)

If the question...	Do this	Where it was asked
Design a Miller two-stage from specs	$C_c = 0.22CL \rightarrow I_5 = SR \cdot C_c \rightarrow gm_1 = 2\pi GB \cdot C_c \rightarrow (W/L)_1 \rightarrow I_{CMR+} \rightarrow (W/L)_3 \rightarrow I_{CMR-} \rightarrow (W/L)_5 \rightarrow gm_7 = 10gm_1 \rightarrow (W/L)_7 \rightarrow (W/L)_8$.	Mid-sem 2025-26 Q2 (design a Miller-compensated two-stage op amp)
Cc vs CL for a given PM	$PM = 90^\circ - \tan^{-1}(\omega_u/\omega_{p2}) - \tan^{-1}(\omega_u/\omega_z)$ with $\omega_u = gm_1/C_c$, $\omega_{p2} = gm_2/CL$, $\omega_z = gm_2/C_c$.	Mid-sem 2023-24 Q5 (peaking factor K at PM 50°; Cc vs CL for 45°) Razavi Example 10.6 (first estimate of CC for 45°) Problem Set 2 P4 (add a second stage to your exam OTA and compensate it)
Dominant-pole compensation	New pole = (crossover you want)/(βA_0); capacitance factor = old pole / new pole.	Past tutorial 2024-25 T2 Ex 5–6 (dominant-pole compensation for closed-loop gains down to 20 dB) Problem Set 2 P3 (compensate the Lec 17 three-pole amplifier)